Talent

Progress Report 3 :
HLD & Literature Review

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COMP0067 App Engineering

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Department of Computer Science

1. Related Technologies Review
1.1 Front-End JavaScript Framework/Library

Our project leverages React to ensure parity with the pre-existing IXN codebase we will extend. Our proposed path forward will avoid the need for refactoring previously deployed code. React is also independently strong: it has the largest developer ecosystem, the best TypeScript support, and direct compatibility with Next.js and Material UI. Angular is too complex for the team's ramp-up timeline.

Risk: Discussions with Dr Daniel Hajas and TSG (UCL IT Services) surfaced a potential risk to proceeding with Next.js 16 - TSG committed to enabling NodeJS 16, which has protocols requiring older, unsupported Next.js versions. While the technical fix is minimal, TSG's dependencies are unknown. We've reached out to confirm this item with TSG Feb-26, and received advice from Dr. Yun Feb-27 that the risk levels for this item are low.

Criterion	React	Vue	Angular
TypeScript support	Native, mature	Good	Native (opinionated)
Server-Side Rendering solution	Next.js	Nuxt.js	Angular Universal
Learning curve	Moderate	Low	High (RxJS, DI)
IXN codebase fit	Existing	Full rewrite	Full rewrite
State of JS 2024	Highest usage	High retention	Declining

Choice: React. Only viable option given the IXN extension constraint; the strongest independent justification on the ecosystem and TypeScript depth.

1.2 Full-Stack React Framework

Next.js is already in the IXN codebase and supports the service-layer architecture planned for this project. Remix performs better for complex, form-heavy applications, but requires a full migration and offers no advantage for this project's main workloads: consent-filtered search and CV retrieval.
Key: RSC = React Server Components; ISR = Incremental Static Regeneration; PE = Progressive Enhancement.

Criterion	Next.js 16	Remix
Rendering	SSR, SSG, ISR, RSC	SSR-first, PE
Data fetching	Server Actions / `getServerSideProps`	Route loaders (web-standard)
IXN codebase fit	Already in use	Full migration
Self-hosted	Full support	Full support

Adoption (2025)	Dominant	Growing

Choice: Next.js 16. Already embedded in the IXN codebase, Remix's performance advantages do not apply to this project's use case.

1.3 React UI Library

The project uses the Mantis dashboard template, which is built on Material UI v5. Ant Design and React Bootstrap are not compatible with Mantis. Material UI provides better capabilities for WCAG (Web Content Accessibility Guidelines) 2.1 AA compliance, which is a directional target for our project.

Criterion	Material UI v5	Ant Design	React Bootstrap
WCAG 2.1 AA	Strong	Moderate	Basic
Mantis template	Native	Incompatible	Incompatible
Component count	~100+	~60+	~40
Theming	sx prop + theme tokens	Less flexible	CSS overrides
Bundle size (gzip)	~80 KB	~140 KB	~25 KB

Choice: Material UI v5. Required by Mantis template integration, best accessibility support, and the largest component library.

1.4 JavaScript ORM

Prisma automatically generates TypeScript types from its schema file, enabling the database schema and application code to stay in sync without manual effort. TypeORM also supports TypeScript but requires manual type definitions. Sequelize is primarily a JavaScript library with partial TypeScript support added later - unsuitable for this project's TypeScript requirement.
Key: ORM = Object-Relational Mapper; SDL = Schema Definition Language.

Criterion	Prisma	TypeORM	Sequelize
TypeScript types	Auto-generated from schema	Manual (decorators)	Partial, JS-first
Migrations	Declarative (`prisma migrate`)	Code-based	Code-based

PostgreSQL support	Full	Full	Full
Query API	Fluent, fully type-safe	Query builder	Verbose, OOP-heavy
Visual tooling	Prisma Studio (GUI)	None	None
Community (2025)	Fastest growth	Stable	Declining

Choice: Prisma. Auto-generated types prevent schema drift; a single schema file reduces bugs; Prisma Studio speeds up debugging for the team.

1.5 Database

PostgreSQL offers a schema aligned with the structured entity relationships for the Talent Platform. MongoDB lacks foreign key constraints, making consent rules harder to enforce. Firebase introduces vendor lock-in and GDPR data residency risk. SQLite cannot handle concurrent writes and is development-only. Redis is a cache layer, not a primary database. Neo4j is graph-oriented and adds unnecessary complexity for this relational data model.
Key: RDBMS = Relational Database; ACID = Atomicity/Consistency/Isolation/Durability

Database	Type	ACID	Full text search	Notes
PostgreSQL	RDBMS	Full	In Built	Native arrays; best Prisma support
MySQL	RDBMS	Full	Limited	No native arrays; weaker FTS
SQLite	RDBMS	Full	Basic	No concurrent writes
MongoDB	Document	Partial	Atlas	No FK constraints
Firebase	Document	Partial	Limited	Vendor lock-in; GDPR risk
Redis	Key-Value	(AOF) Append Only File	Not available	Cache only; not a primary DB
Neo4j	Graph	Full	Limited	Required for highly complex systems only.

Choice: PostgreSQL. available on UCL IXN servers.

1.6 Web Hosting Provider

GDPR and UCL data policy will be handled by the Strategic Alliances Team/UCL IT post-project completion, and hosting via UCL's servers enables this component. External hosting would require UCL Legal sign-off, which is not in scope for V1. The UCL IXN server is the only option with guaranteed UK data residency, no procurement overhead, and an existing deployment path. AWS, Azure, and GCP are over-engineered for 1,000 users at V1 scale. Vercel's serverless model introduces cold start latency (a problem for the mobile-first accessibility goal). IONOS is not UCL-standard infrastructure.

Choice: UCL IXN server. The only option to enable future compliance requirements provides an existing deployment path, and no additional cost.

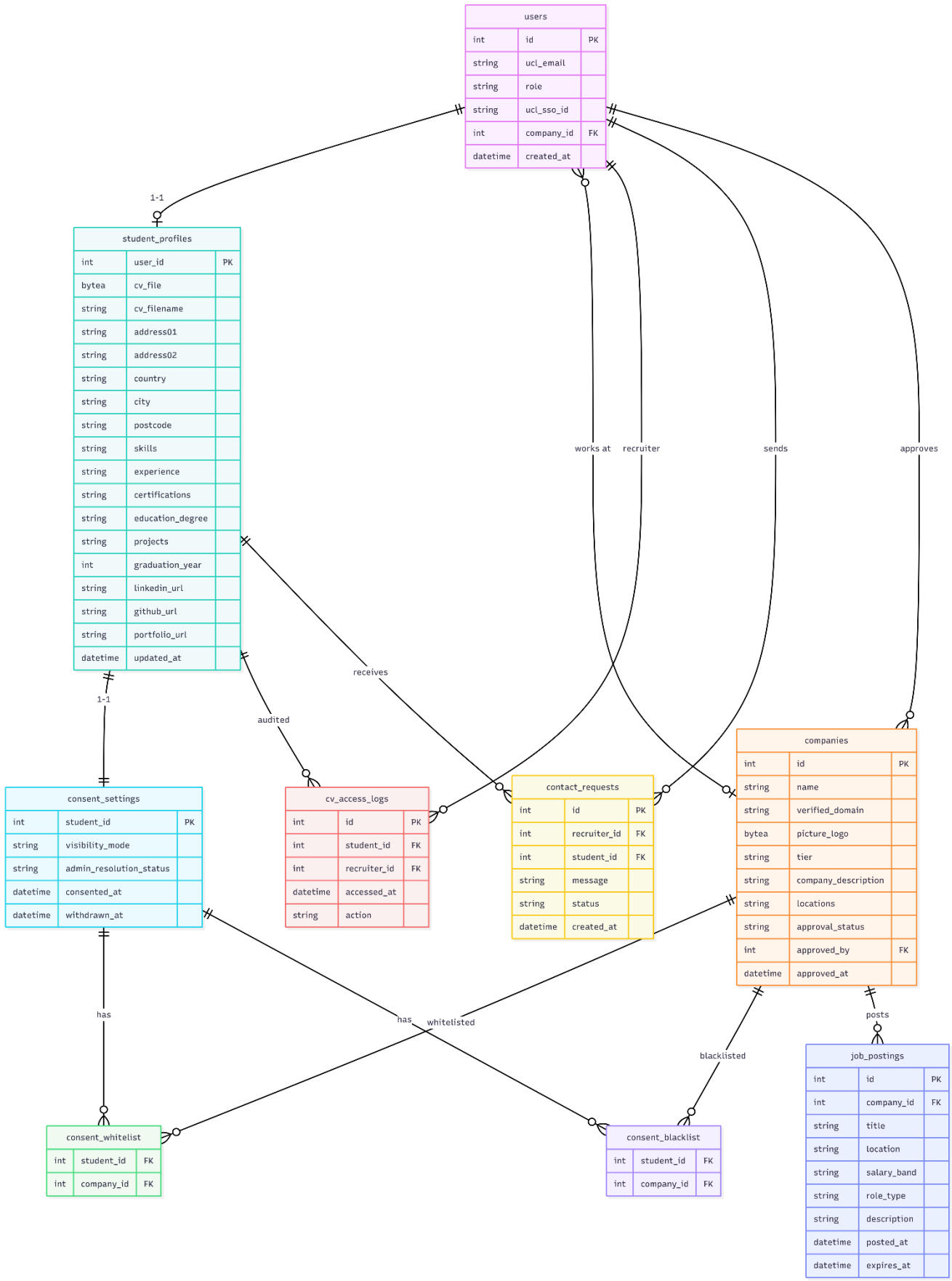
2. Summary of Decisions

The chosen stack is: React (via Next.js 16), TypeScript strict mode, Material UI v5 (Mantis template), Prisma ORM, PostgreSQL, deployed to UCL IXN servers. Every choice extends the existing IXN codebase rather than introducing new platforms, reducing integration risk. Prisma ensures the frontend and backend share one type contract at compile time. PostgreSQL compatibility with the partner's server still requires formal confirmation (raised 29 Jan 2026).
The stack covers all V1 Must Have/Should Have requirements from Report 1: SSO auth (FR-S-1), CV upload (FR-S-4), consent controls (FR-S-12/13), recruiter search (FR-R-3), job postings (FR-R-14), admin approval (FR-A-1), WCAG 2.1 AA (NFR-5), and data residency compliance (NFR-9/11). Features marked Could Have or No in the V1 Build column are excluded from this design.

3. Database Design

The schema has 9 tables mapped directly to V1 functional requirements. Key relationships: `users` 1:1 `student_profiles`; `users` M:1 `companies` (for recruiter accounts); `consent_settings` fans out to `consent_whitelist` and `consent_blacklist` junction tables; `cv_access_logs` is append-only for GDPR audit trail (FR-S-3); `contact_requests` links recruiters to students by email (FR-R-8). The `skills` column is a PostgreSQL `text[]` array with a GIN index for search performance (FR-R-3).

Out-of-scope tables intentionally excluded: no analytics tables (FR-A-3 - Could Have), no saved candidates (FR-R-5), no messaging history (FR-R-9 - email only for V1), no multi-user recruiter access (FR-R-2).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string country
string city
string postcode
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string experience
string certifications
string education_degree
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approves
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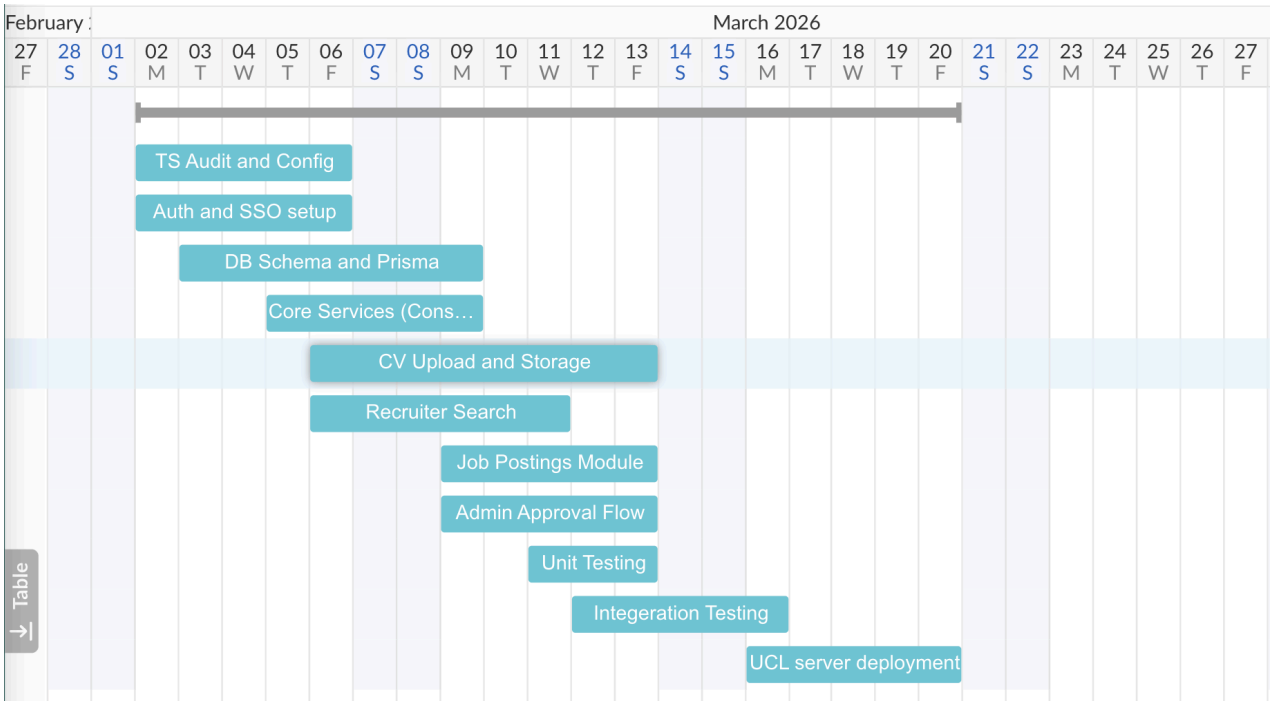
4. Project Schedule

The Gantt spans Feb-May 2026 across five tracks. Coding begins with TypeScript audit (23 Feb-6 Mar), then auth/SSO setup (2-13 Mar) and database schema/Prisma (7-20 Mar) in parallel. Core consent services build through 3 Apr; CV upload, recruiter search, job postings, and admin approval run in overlapping sprints through 1 May. Unit and integration testing overlap from 18 Apr to 15 May. UCL server deployment runs 2-22 May. Progress Report 4 targets 26 Apr-9 May; final report 10-24 May; demo video 17-31 May.

The critical path is: Auth/SSO - Consent services - Recruiter search (11 weeks). CV upload and job postings can proceed in parallel once the DB schema is completed.

5. GANT CHART



6. Project Management

6.1 Schedule Assessment

The project is on schedule. Technology decisions are finalised, and the database schema is complete. The TypeScript audit begins in Week 1 of March as planned. Auth/SSO integration reuses existing IXN components, reducing risk on the critical path. PostgreSQL provisioning confirmation is still outstanding, but it does not block immediate TypeScript config work.

6.2 Partner Assistance Required

1. PostgreSQL provisioning - UCL IXN server setup timeline and credentials needed by 6 Mar to unblock Prisma schema migrations.
2. UCL SSO endpoints - Shibboleth/SAML metadata URL and UCL IT technical contact needed by 13 Mar for auth integration testing.

6.3 Team Contributions (Past Two Weeks)

- Sadhana Jayakumar: Completed the Gantt chart and helped with report lookover.
- Adam Benmohamed: Fixing up the overall look of Figma pages, helping with report lookover.
- Lienqi Peng: helping with ER Diagram.
- Chengyu Yang: Improve the ER diagram, complete the report and modify the Figma as required.
- Gian Prem Rajaram: Supported report 3 review.

6.4 Partner Meetings

- 12/02/2025: Discussed additional requirements to enable administrators to recommend students directly to companies, and reviewed previously completed work on the project with Marco to ensure alignment and continuity.
- 26/02/205: Discussed integration planning and potential challenges when integrating with the existing scaffold project, including requirement alignment, version compatibility issues, necessary adjustments to ensure components work together, and deployment considerations.:

6.5 Plan for Next Two Weeks (27 Feb - 13 Mar)

During the first week (27 Feb – 6 Mar), the focus will be on establishing the system foundation. This includes completing the technical stack audit and configuration, implementing authentication and SSO with role-based access control, finalising the database schema with Prisma migrations, and setting up the core services architecture with standardised error handling and API structure.

In the second week (7 Mar – 13 Mar), development will shift to feature implementation and validation. Key modules include CV upload and storage, recruiter search functionality, job postings management, and the admin approval workflow. Unit testing will be conducted across core services to ensure stability, preparing the system for the upcoming integration testing phase.